CALIBRATION SHEETS

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Calibration Manual Generation Date: 18 March 2004

SBE 37-SM MicroCAT

Conductivity and Temperature Recorder with RS-232 Interface

Instrument Configuration:

Serial Number:	37SM35514-3383
Firmware Version:	2.5
Memory:	2048K bytes
Conductivity Range:	0-7 S/m
Pressure Sensor:	none installed
Interface type:	RS-232
Baud Rate:	9600
External Power Option:	not installed
Maximum Depth:	**7000 meters**

CAUTION - The maximum deployment depth will be limited by the measurement range of the optional pressure sensor, if installed

Sea-Bird Electronics, Inc.
1808 136th Place NE, Bellevue, Washington 98005 USA
Website: http://www.seabird.com

Phone: (425) 643-9866
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Email: seabird@seabird.com

SBE Pressure Test Certificate

Test Date: **11/24/2003** Description **SBE-37 Microcat**

Job Number: **35514** Customer Name **CAN F&O/MLI**

SBE Sensor Information:

Model Number: **37**
Serial Number: **3383**

Pressure Sensor Information:

Sensor Type: **None**
Sensor Serial Number: **None**
Sensor Rating: **0**

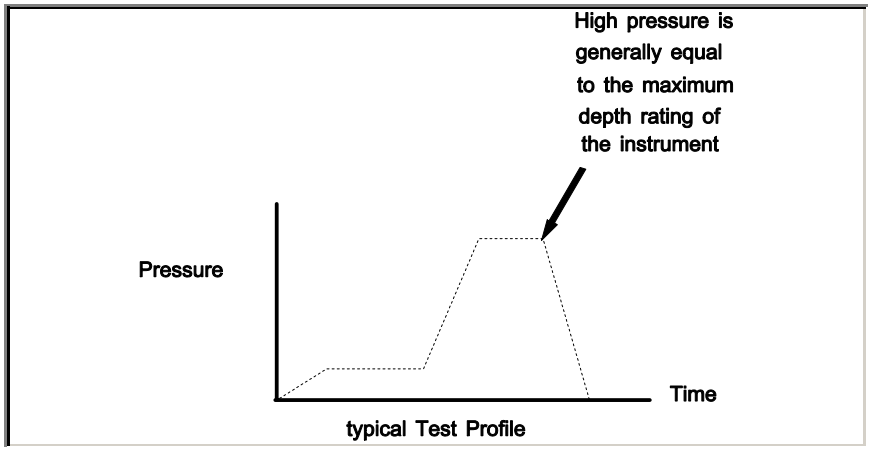
Pressure Test Protocol:

Low Pressure Test: **50** PSI Held For **15** Minutes

High Pressure Test: **10000** PSI Held For **15** Minutes

Passed Test: ☑

Tested By: PCC



typical Test Profile

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1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 3383
CALIBRATION DATE: 15-Mar-04

SBE 37 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

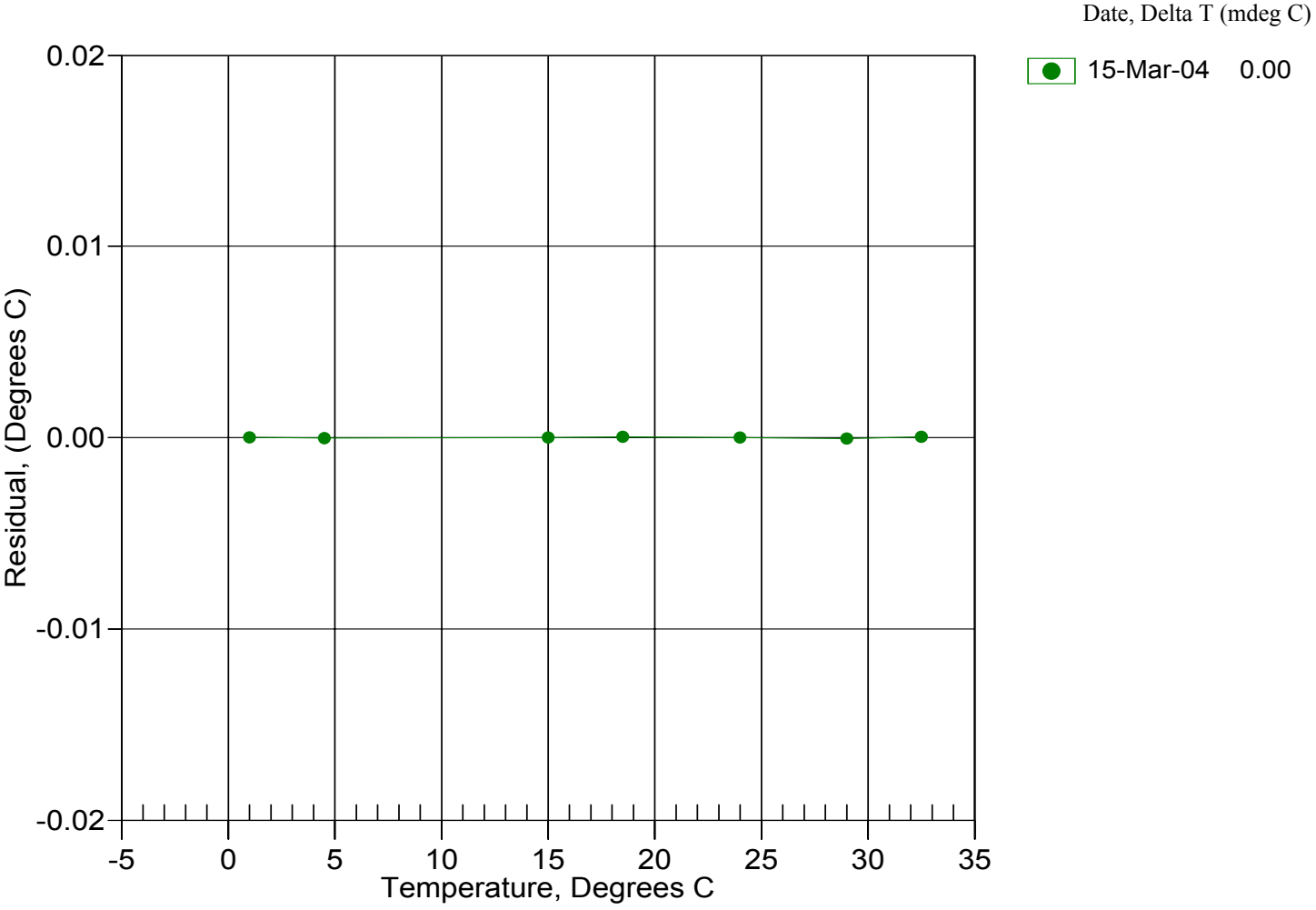
ITS-90 COEFFICIENTS

a0 = -5.046655e-006
a1 = 2.767399e-004
a2 = -2.390979e-006
a3 = 1.567234e-007

BATH TEMP (ITS-90)	INSTRUMENT OUTPUT	INST TEMP (ITS-90)	RESIDUAL (ITS-90)
1.0000	652493.1	1.0000	0.0000
4.5000	558730.2	4.5000	-0.0000
14.9999	357611.4	14.9999	-0.0000
18.4999	310069.5	18.4999	0.0000
24.0000	249252.1	24.0000	-0.0000
28.9999	205602.9	28.9998	-0.0001
32.5000	180264.3	32.5000	0.0000

Temperature ITS-90 = $1/\{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]\} - 273.15$ (°C)

Residual = instrument temperature - bath temperature



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SENSOR SERIAL NUMBER: 3383
CALIBRATION DATE: 15-Mar-04

SBE 37 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -9.833519e-001
h =  1.360156e-001
i = -1.404433e-004
j =  3.071029e-005

CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC = -6.9056e-006
```

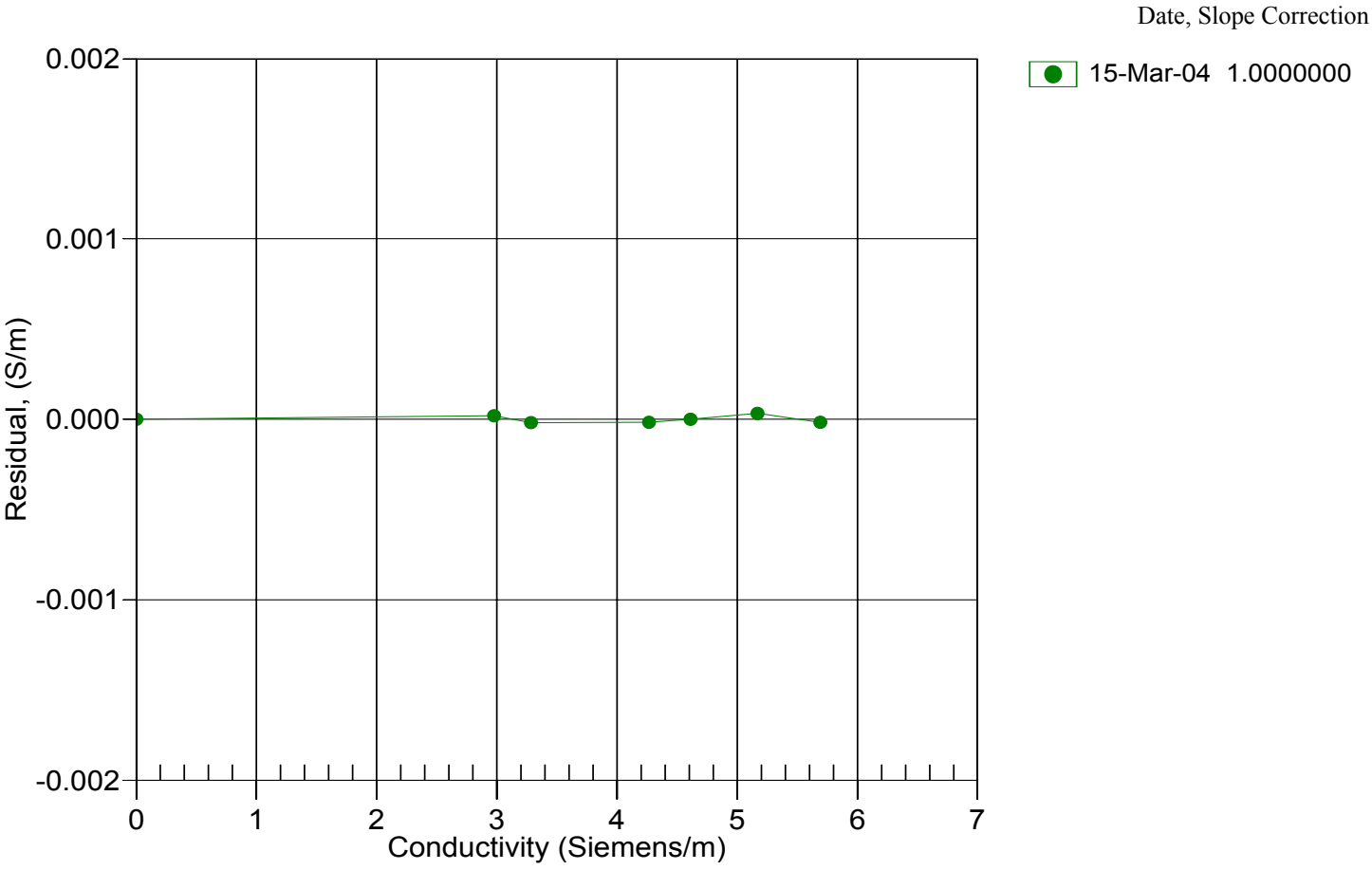
BATH TEMP (ITS-90)	BATH SAL (PSU)	BATH COND (Siemens/m)	INST FREQ (Hz)	INST COND (Siemens/m)	RESIDUAL (Siemens/m)
22.0000	0.0000	0.00000	2690.55	-0.00000	-0.00000
1.0000	34.8330	2.97726	5393.53	2.97728	0.00002
4.5000	34.8123	3.28438	5597.98	3.28436	-0.00002
14.9999	34.7674	4.26624	6205.87	4.26622	-0.00002
18.4999	34.7576	4.61141	6405.61	4.61141	0.00000
24.0000	34.7465	5.16939	6715.64	5.16942	0.00003
28.9999	34.7400	5.69122	6992.78	5.69120	-0.00002

f = INST FREQ * sqrt(1.0 + WBOTC * t) / 1000.0

Conductivity = $(g + hf^2 + if^3 + jf^4) / (1 + \delta t + \varepsilon p)$ Siemens/meter

t = temperature[°C)]; p = pressure[decibars]; δ = CTcor; ε = CPcor;

Residual = instrument conductivity - bath conductivity



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SENSOR SERIAL NUMBER: 3383
CALIBRATION DATE: 15-Mar-04

SBE 37 RTC CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

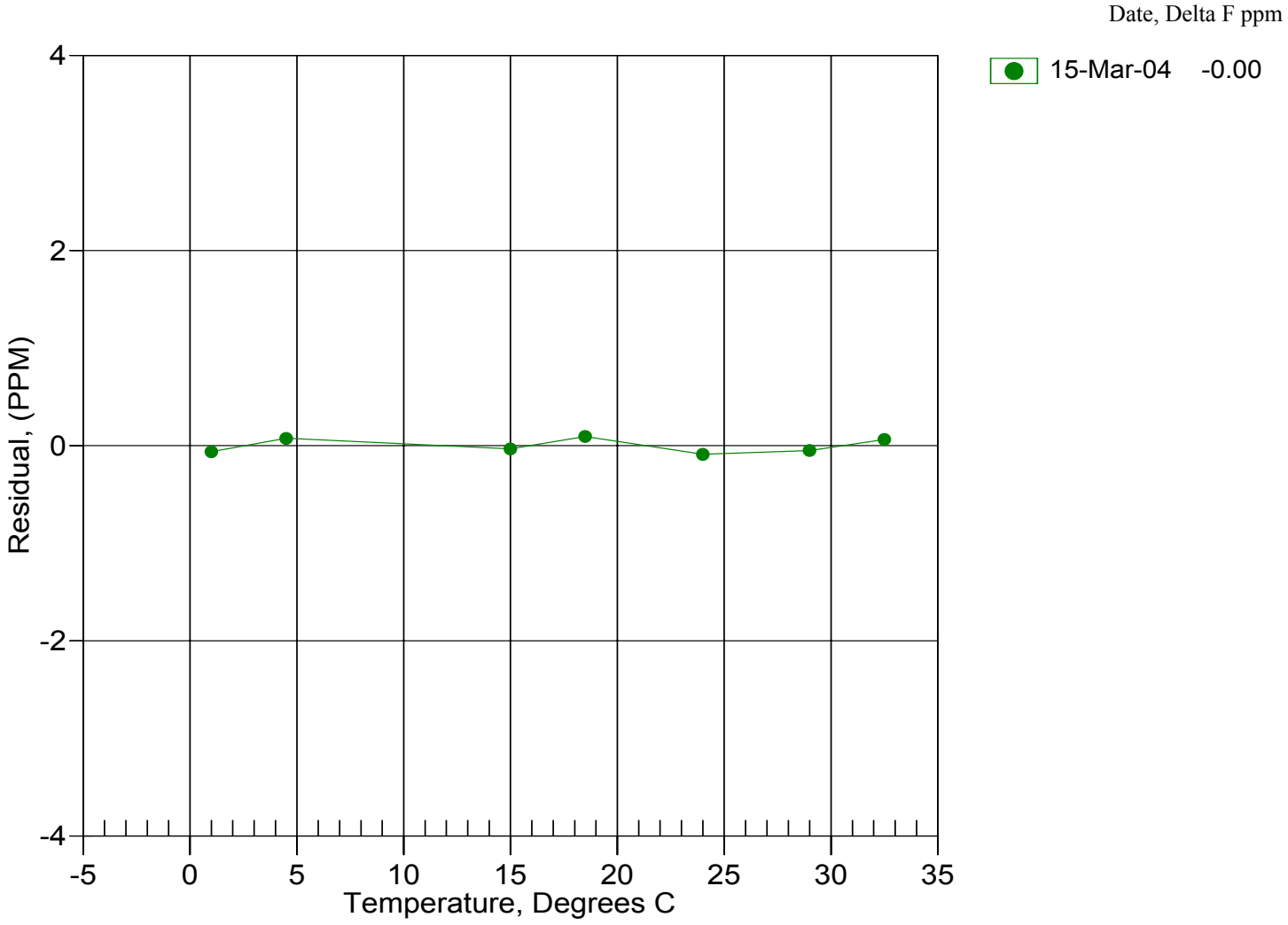
COEFFICIENTS:

rtca0 = 9.999763e-001
rtca1 = 1.636631e-006
rtca2 = -3.073138e-008

BATH TEMP (ITS-90)	RTC FREQ (Hz)	COMPUTED FREQ (Hz)	RESIDUAL (PPM)
1.0000	0.9999780	0.9999779	-0.1
4.5000	0.9999830	0.9999831	0.1
14.9999	0.9999940	0.9999940	-0.0
18.4999	0.9999960	0.9999961	0.1
24.0000	0.9999980	0.9999979	-0.1
28.9999	0.9999980	0.9999980	-0.0
32.5000	0.9999970	0.9999971	0.1

RTC frequency = rtca0 + rtca1 * t + rtca2 * t^2

Residual = (Computed RTC frequency - Measured RTC frequency) * 1e6



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